

Research Paper

The synergistic effect in the heteronuclear double-atom catalysts on CO activation: Insights from DFT calculations

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ABSTRACT

In order to identify the synergistic effect in the heteronuclear double-atom catalysts (DACs) on CO activation, the homonuclear and heteronuclear DACs have been designed based on graphene supporter. And the three CO dissociation pathways have been considered using the density functional theory (DFT) calculation. It reveals that the heteronuclear DACs (MoX@NC) possesses higher catalytic ability than the homonuclear DACs (2X@NC) for CO activation. And the unpaired *d*-orbital electron plays a significant role on CO activation, which can change the magnetic moment of the DACs. The coordination atoms offer similar local coordination environment for the homonuclear and heteronuclear DACs. However, the Bader charge displays that there is a strong electronic interaction between Mo atom and other transition metals of dimer for the heteronuclear DACs. The synergistic effect of the heteronuclear DACs originates from the enhancement of the intermetallic interaction between the Mo atom and other transition metals. The work offers rational guideline for the synthesis of DACs with excellent catalytic performances, and has great potential for application in various catalytic fields.

1. Introduction

The conversion from synthesis gas (H₂ and CO) to hydrocarbon can be accomplished through the Fischer-Tropsch technology, and using this technology can effectively alleviate energy and environmental issues. And the importance of Fischer-Tropsch synthesis (FTS) have received increasing concern in recent years. Carbon monoxide (CO) activity has attracted great interest, since it is a benchmark reaction in the FTS technology [1–3]. CO activation is mainly composed of three pathways [4–7]: (1) the direct dissociation pathway; (2) H-assisted dissociation pathway via CHO; (3) H-assisted dissociation pathway via COH.

Since first reported in 2011 (using Pt doped on the surface of FeO_x) [8], the single-atom Catalysts (SACs), which stand for highest efficient utilization of atom, have sparked great interest recently [9,10]. SACs are commonly supported on typical two-dimensional (2D) materials [11–14]. The graphene is a suitable candidate for the supporter due to its outstanding properties, and N modification could improve the catalytic performance of graphene-based materials [15–17]. However, the main drawback of SACs is its lack of synergistic active site for the complex catalytic reactions. The double-atom catalysts (DACs), which introduce the dual active site, appears as a new perspective in the heterogeneous

catalysis. The dimer on DACs is considered to be active centers for the catalytic reactions [18]. And the synergistic effect of the dimer plays an elemental role on improving their catalytic performance of DACs. Li et al. [19] reported that Cu₂@2SV, MnCu@2SV and NiCu@2SV exhibited promising catalytic performance for the electroreduction of CO₂, and the optimal catalytic products was different owing to the electronic properties of transition metal. CO was the main product for the Cu₂@2SV, MnCu@2SV was selective toward CH₄, but the optimal catalytic product of NiCu@2SV was CH₃OH. In 2019, Zhang et al. [20] proved that the Pt-Ru dimers presented much better catalytic ability for hydrogen evolution reaction (HER), the catalytic activity of Pt-Ru dimers was 50 times more than that of Pt/C catalysts, and it attributed to a synergy effect generated from the Pt-Ru bonding configuration. Li et al. [21] reported that atomic Co/Ni active sites possessed higher catalytic activity than single noble-metal-free catalysts for oxygen reduction reaction (OER), and the improvement of catalytic activity originated from the synergetic effect of atomic Co/Ni-N-C bonds. Grafting a Pt atom into the Fe-N₄ center could greatly improve catalytic activities for oxygen evolution reaction (OER). The reason was due to the synergy in the new active moiety (Pt₁-O₂-Fe₁-N₄), and it promoted the proton adsorption and reduction kinetics [22]. For electrocatalytic nitrogen reduction

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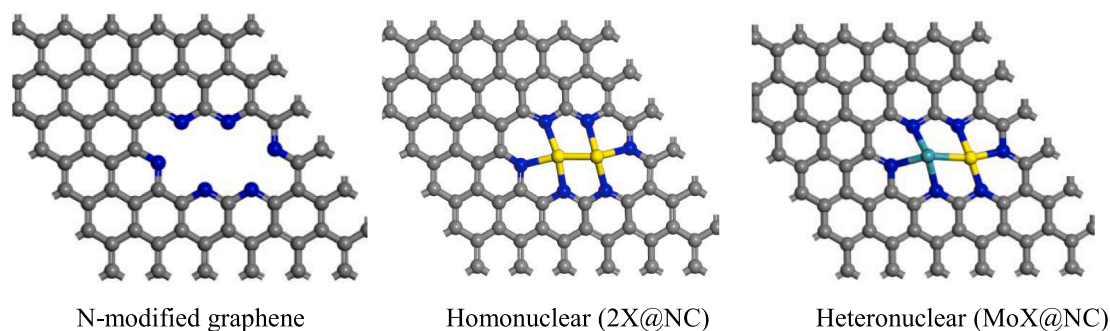


Fig. 1. The configuration of N-modified graphene, homonuclear DACs (2X@NC) and heteronuclear DACs (MoX@NC). X corresponds to 3d or 4d transition metal atoms. (shy ball: C atom; blue ball: N atom; yellow ball: 3d or 4d transition metal atom; cyan ball: Mo atom).

reaction (NRR), the FeMoN₃C dimers revealed higher catalytic activity than its counterparts (Fe@NG and Mo@NG), and it was mainly due to synergistic effects weakening the N≡N bond [23]. For oxygen reduction reaction (ORR) catalytic performance, the Fe-Co active site was better than commercial Pt/C, and the synergistic effects, which rooted from the electronic property of Fe-Co dimer, promoted O=O activation [24]. The Pt-Ru dimer owned better catalytic performance for CO oxidation than the Ru-Ru/Pt-Pt monomers, and the synergetic effect of Pt-Ru dimer balanced the energy evolution of elementary reaction in CO oxidation [25]. It was concluded that the synergistic effect in heteronuclear DACs is a major element of improving the catalytic activity for various reactions. But it is unclear that the synergistic effects in the heteronuclear DACs on CO activation.

In this work, CO activation on the homonuclear and heteronuclear DACs (two 3d- and 4d-transition metals doped on the N-modified graphene materials) will be considered. The stability of the homonuclear and heteronuclear DACs will be evaluated. In view of the activation barriers E_a of carbon monoxide activation, the catalytic ability of the homonuclear and heteronuclear DACs will be analyzed and sequenced. Based on the Bader charge, the coordination environment and the electron transfer between two transition metals of dimer will be characterized. Furthermore, the metallic bond length of dimer on DACs have been discussed.

2. Computational details

2.1. Computational methods

All calculation was performed by the Dmol³ module in Materials Studio 8.0 in consideration of periodic density functional theory (DFT) method [26,27]. The electronic exchange-correlation potential was described by using the generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof functional (PBE) [28,29]. The effective core potential (ECP) set was carried out for transition metal atom, and an all-electron basis set was applied for non-metallic atom [30,31]. The double-numerical basis set with polarization functions (DNP) was employed to treat the valence electron functions [32]. A Monkhorst-Pack grid of $3 \times 3 \times 1$ and a Methfessel-Paxton smearing of 0.005 Ha have been set. Spin-unrestricted calculation was chosen to use different orbitals for different spins. DFT-D correction method was applied for correcting van der Waals (vdW) interaction [33,34]. The transition state was searched by using the complete LST/QST method, and it was confirmed by frequency analysis and TS Confirmation [35].

The average binding energies ($E_{\text{average BE}}$) and the average cohesive energies ($E_{\text{average coh}}$) of the double TM atom on N-modified graphene were calculated as:

$$E_{\text{average BE}} = (E_{\text{M1M2+substrate}} - E_{\text{M1atom}} - E_{\text{M2atom}} - E_{\text{substrate}})/2 \quad (1)$$

$$E_{\text{average coh}} = (E_{\text{coh-M1}} + E_{\text{coh-M2}})/2 \quad (2)$$

where $E_{\text{M1M2+substrate}}$, $E_{\text{M1 atom}}$, $E_{\text{M2 atom}}$, and $E_{\text{substrate}}$ are the energies of double-atom catalysts (DACs), the energies of M₁ atom, the energies of M₂ atom and the total energies of the substrate, respectively. $E_{\text{coh-M1}}$ and $E_{\text{coh-M2}}$ are the cohesive energies of M₁ and M₂ atom, the cohesive energies of transition metal atoms are obtained by Kittel et al. [36].

The adsorption energy (E_{ads}) of related species in the CO activation were defined as:

$$E_{\text{ads}} = E_{\text{catalyst}} + E_{\text{adsorbate}} - E_{\text{total}} \quad (3)$$

where E_{catalyst} , $E_{\text{adsorbate}}$, and E_{total} denotes the energies of the homonuclear and heteronuclear DACs, the energies of adsorbed C₁ molecule, and the total energies of the catalyst with adsorbed species, respectively.

The activation energy (E_a) and reaction energy (ΔH) of the elementary reaction were defined as (1) and (2):

$$E_a = E_{\text{TS}} - E_{\text{IS}} \quad (4)$$

$$\Delta H = E_{\text{FS}} - E_{\text{IS}} \quad (5)$$

where E_{IS} , E_{TS} , and E_{FS} corresponds to the total energy of the initial, transition and final state structure.

2.2. Computational model

In view of the 6×6 supercell of graphene supporter, four carbon atoms have been deleted and the six carbon atoms around the vacancy have been replaced by nitrogen atoms, thus the N-modified graphene have been formed (as seen in Fig. 1). The homonuclear DACs were constructed by doping two 3d- and 4d-transition metal atoms in the N-modified graphene supporter. The heteronuclear DACs were constructed by doping Mo atom and other 3d- and 4d-transition metal atoms in the N-modified graphene supporter. The vacuum of the 15 Å thickness is built for the avoidance of the interaction between the neighboring slabs.

3. Results and discussion

3.1. The adsorption of the C₁ species and H₂ dissociation

The configuration of the homonuclear and heteronuclear DACs are displayed in Figs. S1 and S2. And the absorption energy of CO, CHO and COH on the homonuclear and heteronuclear DACs, the corresponding adsorption configuration are displayed in Table S1–3 and Figs. S3–8. The absorption energy of the CO, CHO and COH over the homonuclear and heteronuclear DACs are greater than 0.73, 1.49 and 2.13 eV, respectively. It indicates that the C₁ small species can be stably adsorbed on the homonuclear and heteronuclear DACs.

The CO activation is comprised of two H-assisted activation mechanism (via CHO and COH), and the H₂ dissociation is the main source of H atom for H-assisted CO activation mechanism. Accordingly, H₂ dissociation on the homonuclear and heteronuclear DACs have been

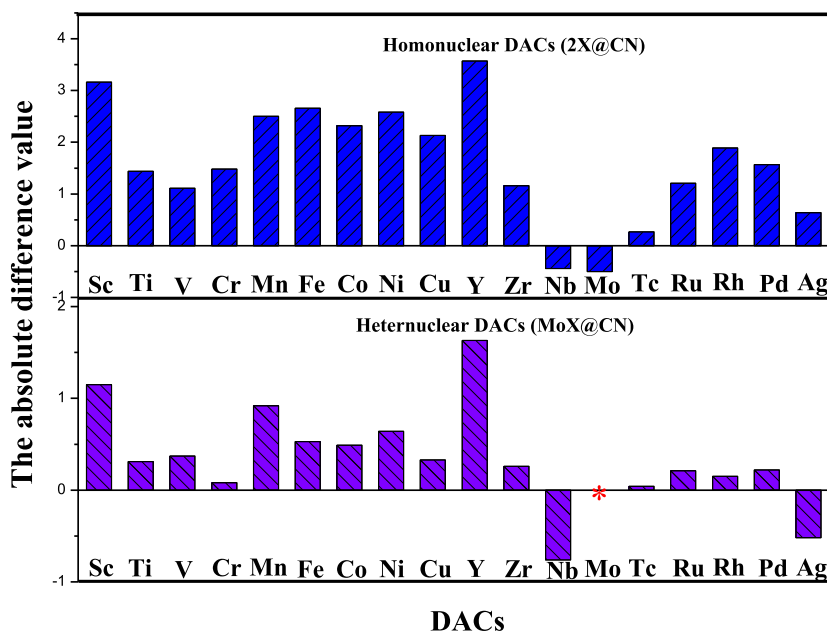


Fig. 2. The difference in absolute value between the average binding energies ($E_{\text{average BE}}$) and the average cohesive energies ($E_{\text{average coh}}$).

Table 1

The activation energy (E_a , eV), the reaction energy (ΔH , eV) and the C=O distance ($d_{\text{C=O}}$, Å) at the transition state of $\text{CO}=\text{C}+\text{O}$ for the homonuclear DACs.

Homonuclear DACs		CO→C+O		
		E_a	ΔH	$d_{\text{C=O}}$
3d-TM	2Sc@NC	3.79	1.41	2.68
	2Ti@NC	2.12	1.11	2.08
	2V@NC	2.16	0.55	1.92
	2Cr@NC	2.25	1.54	1.95
	2Mn@NC	3.16	2.69	1.94
	2Fe@NC	4.17	4.15	2.31
	2Co@NC	6.05	5.51	2.31
	2Ni@NC	6.71	5.91	2.23
4d-TM	2Cu@NC	7.74	7.12	2.27
	2Y@NC	4.05	1.69	2.80
	2Zr@NC	2.20	1.05	2.08
	2Nb@NC	2.19	0.27	2.18
	2Mo@NC	1.52	0.97	1.88
	2Tc@NC	3.01	1.65	1.73
	2Ru@NC	3.67	2.85	1.81
	2Rh@NC	6.33	5.34	2.22
	2Pd@NC	6.72	5.44	2.28
	2Ag@NC	8.66	7.44	2.68

considered. For the homonuclear DACs, H_2 can spontaneous dissociation except for 2Mn@NC, 2Co@NC, 2Ni@NC, 2Cu@NC, 2Pd@NC and 2Ag@NC catalysts. The H_2 dissociation activation energy of 2Mn@NC, 2Co@NC, 2Ni@NC, 2Cu@NC, 2Pd@NC and 2Ag@NC catalysts are 1.21, 1.15, 1.93, 1.89, 1.55 and 0.50 eV, respectively (as presented in Fig. S9). However, H_2 can spontaneous dissociation for all the heteronuclear DACs (as displayed in Fig. S10). Consequently, Hydrogen is easily dissociated for all of the homonuclear and heteronuclear DACs, so they can offer adequate hydrogen atom for H-assisted CO activation mechanism.

3.2. The stability of DACs

Stability is attached great importance to the construction of DACs. In theoretical research, the following method has been used for the sake of discussing the stability of the SACs [37,38] or DACs [39,40]: compared the binding energies of metal on the substrate with the cohesive energies

of metal atom in its bulk form. If the absolute value of average binding energies is larger than that of the cohesive energies for transition metal, it indicates that the SACs or DACs is stable and difficult to assemble into cluster. Consequently, the average binding energies ($E_{\text{average BE}}$) of DACs and the average cohesive energies ($E_{\text{average coh}}$) of the two TM atom have been discussed and presented in the Supplementary Material (Table S4). The absolute difference value between them are displayed in Fig. 2. For the great majority of the homonuclear DACs, the absolute difference value between $E_{\text{average BE}}$ and $E_{\text{average coh}}$ is large, which reveals that they can be stably formed and are difficult to cluster. But the stability of 2Nb@NC and 2Mo@NC catalysts are slightly poor, the absolute difference value of them are negative. However, for the heteronuclear DACs, except for MoNb@NC and MoAg@NC catalysts, the stability of the heteronuclear DACs (MoX@NC) is high. Especially, the stability of ScMo@NC, MnMo@NC and YMo@NC catalysts are dramatically elevated. It can be concluded that other transition metals can be used to improve the stability of 2Mo@NC catalyst.

3.3. CO activation on the homonuclear DACs

3.3.1. CO direct dissociation mechanism

The activation energy E_a , the reaction energy ΔH and the C=O distance at the transition state of $\text{CO}=\text{C}+\text{O}$ are listed in Table 1. And the configuration of initial state (IS), transition state (TS) and final state (FS) of $\text{CO}=\text{C}+\text{O}$ for the homonuclear DACs are displayed in Fig. S11.

For the 3d-TM homonuclear DACs, the 2Ti@NC catalyst is the most advantageous for CO activation, followed by 2V@NC catalyst (as revealed in Table 1). The activation energy E_a of 2Ti@NC and 2V@NC catalyst are 2.12 and 2.16 eV, the C=O distance of 2Ti@NC and 2V@NC are 2.08 Å and 1.92 Å at the transition state. However, the activation energy E_a of 2Cu@NC catalyst is 7.74 eV, and it is the greatest among them, and the C=O distance is 2.27 Å at the transition state. Based on the activation energy E_a , it displays the following sequence: $\text{Ti} < \text{V} < \text{Cr} < \text{Mn} < \text{Sc} < \text{Fe} < \text{Co} < \text{Ni} < \text{Cu}$. For the 4d-TM homonuclear DACs, the 2Mo@NC catalyst exhibits the highest CO activation performance. The activation energy E_a of 2Mo@NC catalyst is 1.52 eV, and the C=O distance is 1.88 Å at the transition state. However, the activation energy E_a of 2Ag@NC catalyst is 8.66 eV, which is the greatest among them. And the C=O distance is 2.68 Å at the transition state. Based on the activation energy E_a , it displays the following sequence:

Table 2

The (total) activation energy (E_{a1} ; E_{a2} ; E_{total} , eV) and the reaction energy (ΔH_1 ; ΔH_2 , eV) of the CO dissociation through CHO for the homonuclear DACs.

Homonuclear DACs		CO+H→CHO		CHO→CH+O		CO+H→CH+O
		E_{a1}	ΔH_1	E_{a2}	ΔH_2	E_{total}
3d-TM	2Sc@NC	1.48	-0.61	1.70	-0.47	1.48
	2Ti@NC	1.92	-0.27	1.64	0.59	1.92
	2V@NC	1.31	0.13	1.16	0.72	1.31
	2Cr@NC	0.78	0.15	0.83	0.21	0.98
	2Mn@NC	0.94	0.17	1.50	1.22	1.67
	2Fe@NC	0.88	0.38	2.69	2.59	3.07
	2Co@NC	0.60	0.35	4.71	3.79	5.06
	2Ni@NC	0.83	-0.55	5.79	5.44	5.24
	2Cu@NC	0.88	-0.12	5.88	3.74	5.76
	2Y@NC	1.31	-0.50	2.86	2.01	2.36
4d-TM	2Zr@NC	1.98	-0.19	0.97	-0.63	1.98
	2Nb@NC	1.25	-0.62	0.65	-0.08	1.25
	2Mo@NC	1.28	0.22	0.54	-0.89	1.28
	2Tc@NC	2.09	0.77	1.26	-0.04	2.09
	2Ru@NC	2.21	0.62	2.17	1.91	2.79
	2Rh@NC	1.98	0.68	5.41	4.48	6.09
	2Pd@NC	1.34	-0.01	6.67	5.68	6.66
	2Ag@NC	0.67	0.18	6.65	3.51	6.83

Table 3

The (total) activation energy (E_{a3} ; E_{a4} ; E_{total} , eV) and the reaction energy (ΔH_3 ; ΔH_4 , eV) of the CO activation through COH for the homonuclear DACs.

Homonuclear DACs		CO+H→COH		COH→C+OH		CO+H→C+OH
		E_{a3}	ΔH_3	E_{a4}	ΔH_4	E_{total}
3d-TM	2Sc@NC	3.49	0.80	3.15	0.45	3.95
	2Ti@NC	1.64	0.66	1.40	-1.07	2.06
	2V@NC	2.15	0.39	1.83	0.16	2.22
	2Cr@NC	2.07	0.37	2.36	0.98	2.73
	2Mn@NC	2.33	0.73	2.49	0.90	3.22
	2Fe@NC	2.79	1.00	2.37	1.62	3.37
	2Co@NC	1.82	0.79	3.46	3.39	4.25
	2Ni@NC	3.48	1.27	3.99	2.85	5.26
	2Cu@NC	3.46	1.22	4.15	2.24	5.37
	2Y@NC	2.57	1.10	3.42	0.47	4.52
4d-TM	2Zr@NC	1.66	1.12	2.69	1.35	3.81
	2Nb@NC	2.24	0.67	1.88	-0.24	2.55
	2Mo@NC	2.20	0.54	2.02	0.38	2.56
	2Tc@NC	3.22	0.82	2.90	0.73	3.72
	2Ru@NC	2.04	0.89	2.85	1.50	3.74
	2Rh@NC	1.88	0.90	4.63	4.18	5.53
	2Pd@NC	3.77	1.86	3.48	3.32	5.34
	2Ag@NC	2.83	1.21	5.06	-0.40	6.27

Mo<Nb<Zr<Tc<Ru<Y<Rh<Pd<Ag. The activation energy E_a of 2Mo@NC catalyst is the lowest of all the homonuclear DACs. And at the transition state the C=O distance relates with the activation energy E_a , it can be concluded that at the transition state the smaller the C=O distance, the lower the activation energy of the homonuclear DACs.

3.3.2. H-assisted CO dissociation mechanism through CHO

The activation energy (E_{a1} ; E_{a2}) and the reaction energy (ΔH_1 ; ΔH_2) of CO+H=CHO and CHO=CH+O, and the total activation energy (E_{total}) of CO+H=CH+O are listed in Table 2, the corresponding configuration of initial state (IS), transition state (TS), and final state (FS) of CO+H=CHO=CH+O for the homonuclear DACs are displayed in Fig. S12.

For the 3d-TM homonuclear DACs, the activation energy E_{a1} of CO+H→CHO is in the range of 0.60 to 1.92 eV, it indicates that CHO is easy to generate through CO hydrogenation (As displayed in Table 2). The activation energy E_{a2} of CHO→CH+O is in the range of 0.83 to 5.88 eV. In view of two continuous reaction, the 2Cr@NC catalyst possesses the highest catalytic activity for the CO activation through CHO, the total activation energy E_{total} of 2Cr@NC catalyst is 0.98 eV. However, the total activation energy E_{total} of 2Cu@NC catalyst is 5.76 eV, and it is

the greatest among them. For the 4d-TM homonuclear DACs, the activation energy E_{a1} of CO+H→CHO is in the range of 0.67 to 2.21 eV, it displays that CHO is easy to generate through CO hydrogenation for the 4d-TM homonuclear DACs similarly. The activation energy E_{a2} of CHO→CH+O is in the range of 0.54 to 6.67 eV, it reveals that C=O bond rupture of CHO depend on electronic properties of transition metals. In view of two continuous reaction, the 2Nb@NC catalyst possesses the highest catalytic activity for CO activation through CHO, followed by 2Mo@NC catalyst. The total activation energy E_{total} of 2Nb@NC and 2Mo@NC catalysts are 1.25 and 1.28 eV. However, the total activation energy E_{total} of 2Ag@NC catalyst is 6.83 eV, and it is the greatest among them.

3.3.3. H-assisted CO dissociation mechanism through COH

The activation energy (E_{a3} ; E_{a4}) and the reaction energy (ΔH_3 ; ΔH_4) of CO+H=COH and COH=C+OH, and the total activation energy (E_{total}) of CO+H=C+OH are listed in Table 3, the corresponding configuration of initial state (IS), transition state (TS), and final state (FS) of CO+H=COH=C+OH for the homonuclear DACs are displayed in Fig. S13.

For the 3d-TM homonuclear DACs, the activation energy E_{a3} of CO+H→COH is in the range of 1.64 to 3.49 eV, it indicates that COH is difficult to generate through CO hydrogenation for all the 3d-TM homonuclear DACs (as presented in Table 3). The activation energy E_{a4} of COH→C+OH is in the range of 1.40–4.15 eV. In view of two continuous reaction, the 2Ti@NC catalyst owns the best catalytic performance for the CO activation through COH. The total activation energy E_{total} of 2Ti@NC catalyst is 2.06 eV. However, the total activation energy E_{total} of 2Cu@NC catalyst is 5.37 eV, and it is the greatest among them. For the 4d-TM homonuclear DACs, the activation energy E_{a3} of CO+H→COH is in the range of 1.66 to 3.77 eV, it indicates that COH is difficult to generate through CO hydrogenation for all the 4d-TM homonuclear DACs. Hydrogenation to CHO is the most favorable pathway for carbon monoxide. The activation energy E_{a4} of COH→C+OH is in the range of 1.88 to 5.06 eV. In view of two continuous reaction, 2Nb@NC catalyst owns the best catalytic performance for the CO activation through COH, followed by 2Mo@NC catalyst. The total activation energy E_{total} of 2Nb@NC and 2Mo@NC catalysts are 2.55 and 2.56 eV. However, the total activation energy E_{total} of 2Ag@NC catalyst is 6.27 eV, which is the greatest among them.

For CO small molecule, hydrogenation to CHO is more favorable than hydrogenation to COH, and the C=O bond rupture of CHO is much easier than that of CO molecule. Compared with other pathways, CO+H→CHO→CH+O is the most advantageous for CO activation for the great majority of the homonuclear DACs (except for 2Co@NC, 2Cu@NC, 2Rh@NC, 2Pd@NC and 2Ag@NC catalysts, the optimal path of them is CO+H→COH→C+OH). The 2Cr@NC catalyst is the most advantageous for the H-assisted CO dissociation through CHO, followed by 2Nb@NC and 2Mo@NC catalysts, however, there is only 0.3 eV difference value among them. And 2Mo@NC catalyst possesses the best catalytic performance for C=O bond rupture of CHO than others (the activation energy E_{a2} of 2Mo@NC catalyst is 0.54 eV for CHO→CH+O). Meanwhile, 2Mo@NC catalyst possesses better catalytic ability than others for the remaining CO dissociation pathways, the activation energy E_a of 2Mo@NC catalyst is only 1.52 eV for CO=C+O pathway, which is obvious lower than others. The E_{total} of 2Mo@NC catalyst is 2.56 eV for CO+H=C+OH pathways, which is second only to 2Nb@NC catalysts. Therefore, the 2Mo@NC catalyst is the suitable candidate for CO activation. More importantly, Mo-based SACs or DACs possesses best catalytic ability for electrochemical N₂ reduction and hydrogen evolution reaction in experimental research [41–43]. With regard to the pure transition metal catalysts, the transition metals situated in right of periodic table possess weaker CO activation ability (such as Pd and Cu), the C=O bond does not break and methanol is the main product for the Pd and Cu catalysts [44]. The W and Mo catalysts possess strong CO activation ability and easily form metal carbides [45,46]. It is in agreement

Table 4

The activation energy (E_a , eV), the reaction energy (ΔH , eV) and the C=O distance at the transition state ($d_{c=O}$, Å) of CO→C+O for the heteronuclear DACs.

Heteronuclear DACs.		CO→C+O		
		E_a	ΔH	$d_{c=O}$
3d-TM	MoSc@NC	2.93	1.24	1.96
	MoTi@NC	1.97	1.45	1.93
	MoV@NC	1.57	0.33	1.93
	MoCr@NC	2.09	0.85	1.96
	MoMn@NC	2.96	1.84	1.97
	MoFe@NC	3.47	2.63	1.96
	MoCo@NC	2.91	1.84	1.98
	MoNi@NC	3.67	2.32	2.02
	MoCu@NC	3.43	2.44	2.04
	MoY@NC	3.50	2.08	2.33
4d-TM	MoZr@NC	2.07	0.64	1.93
	MoNb@NC	1.40	-0.25	1.86
	MoTc@NC	2.82	1.83	1.78
	MoRu@NC	3.78	2.27	1.95
	MoRh@NC	4.03	3.81	2.15
	MoPd@NC	4.67	4.27	2.10
	MoAg@NC	2.42	1.84	2.02

Table 5

The (total) activation energy (E_{a1} ; E_{a2} ; E_{total} , eV) and the reaction energy (ΔH_1 ; ΔH_2 , eV) of the CO dissociation through CHO for the heteronuclear DACs.

Heteronuclear DACs.		CO+H→CHO		CHO→CH+O		E_{total}
		E_{a1}	ΔH_1	E_{a2}	ΔH_2	
3d-TM	MoSc@NC	0.82	0.41	1.25	0.51	1.66
	MoTi@NC	0.65	0.03	0.81	-0.37	0.84
	MoV@NC	0.73	-0.17	0.58	-0.36	0.73
	MoCr@NC	1.27	0.92	0.41	-0.29	1.33
	MoMn@NC	1.63	0.81	1.31	0.36	2.12
	MoFe@NC	1.50	0.50	1.37	-0.48	1.87
	MoCo@NC	1.42	1.09	0.99	-0.61	2.08
	MoNi@NC	1.48	0.85	2.35	0.02	3.20
	MoCu@NC	1.79	0.62	1.98	0.46	2.60
	MoY@NC	0.78	0.11	1.87	0.93	1.98
4d-TM	MoZr@NC	0.58	-0.02	1.08	-0.37	1.06
	MoNb@NC	0.39	-0.46	0.64	-0.36	0.39
	MoTc@NC	2.10	1.07	1.43	-0.39	2.50
	MoRu@NC	1.90	0.72	1.44	0.60	2.16
	MoRh@NC	0.38	0.15	1.78	1.40	1.93
	MoPd@NC	0.76	0.14	2.03	0.42	2.17
	MoAg@NC	1.29	0.61	1.95	0.47	2.56

with our results for the homonuclear DACs.

Consequently, the heteronuclear DACs is constructed based on the 2Mo@NC catalyst, one of Mo atom is replaced by other 3d- or 4d-TM atoms in order to form the heteronuclear DACs (MoX@NC).

3.4. CO activation on the heteronuclear DACs

3.4.1. CO direct dissociation mechanism

The activation energy E_a , the reaction energy ΔH and the C=O distance at the transition state of CO→C+O are listed in Table 4. And the configuration of initial state (IS), transition state (TS) and final state (FS) of CO→C+O for the heteronuclear DACs are revealed in Fig. S14.

For the 3d-TM heteronuclear DACs (as presented in Table 4), the MoV@NC catalyst displays the best catalytic ability for CO direct dissociation, the activation energy E_a is 1.57 eV, and the C=O distance of MoV@NC is 1.93 Å at the transition state. However, the activation energy E_a of MoNi@NC catalyst is 3.67 eV, which is the greatest among them, and the C=O distance is 2.02 Å at the transition state. For the 4d-TM heteronuclear DACs, the MoNb@NC catalyst displays the highest catalytic ability for CO activation. The activation energy E_a of MoNb@NC catalyst is 1.40 eV, and the C=O distance is 1.86 Å at the transition state. However, the activation energy E_a of MoPd@NC

Table 6

The (total) activation energy (E_{a3} ; E_{a4} ; E_{total} , eV) and the reaction energy (ΔH_3 ; ΔH_4 , eV) of the CO dissociation through COH for the heteronuclear DACs.

Heteronuclear DACs		CO+H=COH		COH=C+OH		CO+H=C+OH
		E_{a3}	ΔH_3	E_{a4}	ΔH_4	
3d-TM	MoSc@NC	3.81	1.35	0.48	-0.98	3.81
	MoTi@NC	3.72	1.22	0.93	-0.93	3.72
	MoV@NC	2.95	0.67	1.59	0.07	2.95
	MoCr@NC	2.29	0.72	2.15	0.72	2.87
	MoMn@NC	3.14	0.90	2.23	0.62	3.14
	MoFe@NC	3.13	0.77	2.30	1.03	3.13
	MoCo@NC	3.33	0.94	3.23	2.03	4.17
	MoNi@NC	3.52	1.23	2.78	1.68	4.01
	MoCu@NC	3.58	1.34	2.11	0.87	3.58
	MoY@NC	3.84	1.43	0.67	-1.26	3.84
4d-TM	MoZr@NC	3.85	1.38	0.83	-1.43	3.85
	MoNb@NC	2.17	0.62	1.27	-0.46	2.17
	MoTc@NC	3.35	0.78	2.83	1.21	3.61
	MoRu@NC	3.08	0.81	3.28	1.09	4.09
	MoRh@NC	2.77	0.34	3.94	2.55	4.28
	MoPd@NC	2.94	0.64	2.49	1.56	3.13
	MoAg@NC	3.30	0.52	2.89	1.84	3.41

catalyst is 4.67 eV, which is the greatest among them, and the C=O distance is 2.10 Å at the transition state. The activation energy E_a of MoNb@NC catalyst is the lowest of all the homonuclear DACs. It is indicated that at the transition state the smaller the C=O atom distance, the lower the activation energy E_a of the heteronuclear DACs (similar to the homonuclear DACs).

3.4.2. H-assisted CO dissociation mechanism through CHO

The activation energy (E_{a1} ; E_{a2}) and the reaction energy (ΔH_1 ; ΔH_2) of CO+H=CHO and CHO=CH+O, and the total activation energy (E_{total}) of CO+H=CH+O are listed in Table 5, the corresponding configuration of initial state (IS), transition state (TS), and final state (FS) of CO+H=CHO=CH+O for the heteronuclear DACs are displayed in Fig. S15.

For the 3d-TM heteronuclear DACs (as revealed in Table 5), the activation energy E_{a1} of CO+H→CHO is in the range of 0.65 to 1.79 eV, it displays that CHO is easy to generate through CO hydrogenation for the 3d-TM heteronuclear DACs. The activation energy E_{a2} of CHO→CH+O is in the range of 0.41 to 2.35 eV. In view of two continuous reaction, the MoV@NC catalyst possesses the highest catalytic activity for the CO activation through CHO, followed by MoTi@NC catalyst. The total activation energy E_{total} of MoV@NC and MoTi@NC catalysts are 0.73 and 0.84 eV. However, the total activation energy E_{total} of MoNi@NC catalyst is 3.20 eV, and it is the greatest among them. For the 4d-TM heteronuclear DACs, the activation energy E_{a1} of CO+H→CHO is in the range of 0.38–2.10 eV, it indicates that CHO is easy to generate through CO hydrogenation for the 4d-TM heteronuclear DACs similarly. The activation energy E_{a2} of CHO→CH+O is in the range of 0.64 to 2.03 eV. In view of two continuous reaction, the MoNb@NC catalyst possesses the highest catalytic activity for CO activation through CHO, followed by MoZr@NC catalyst. The total activation energy E_{total} of MoNb@NC and MoZr@NC catalysts are 0.39 and 1.06 eV. However, the total activation energy E_{total} of MoAg@NC catalyst is 2.56 eV, and it is the greatest among them.

3.4.3. H-assisted CO dissociation mechanism through COH

The activation energy (E_{a3} ; E_{a4}) and the reaction energy (ΔH_3 ; ΔH_4) of CO+H=COH and COH=C+OH, and the total activation energy (E_{total}) of CO+H=C+OH are listed in Table 6, the corresponding configuration of initial state (IS), transition state (TS), and final state (FS) of CO+H=COH=C+OH for the heteronuclear DACs are displayed in Fig. S16.

For the 3d-TM heteronuclear DACs (as presented in Table 6), the activation energy E_{a3} of CO+H→COH is in the range of 2.29 to 3.81 eV,

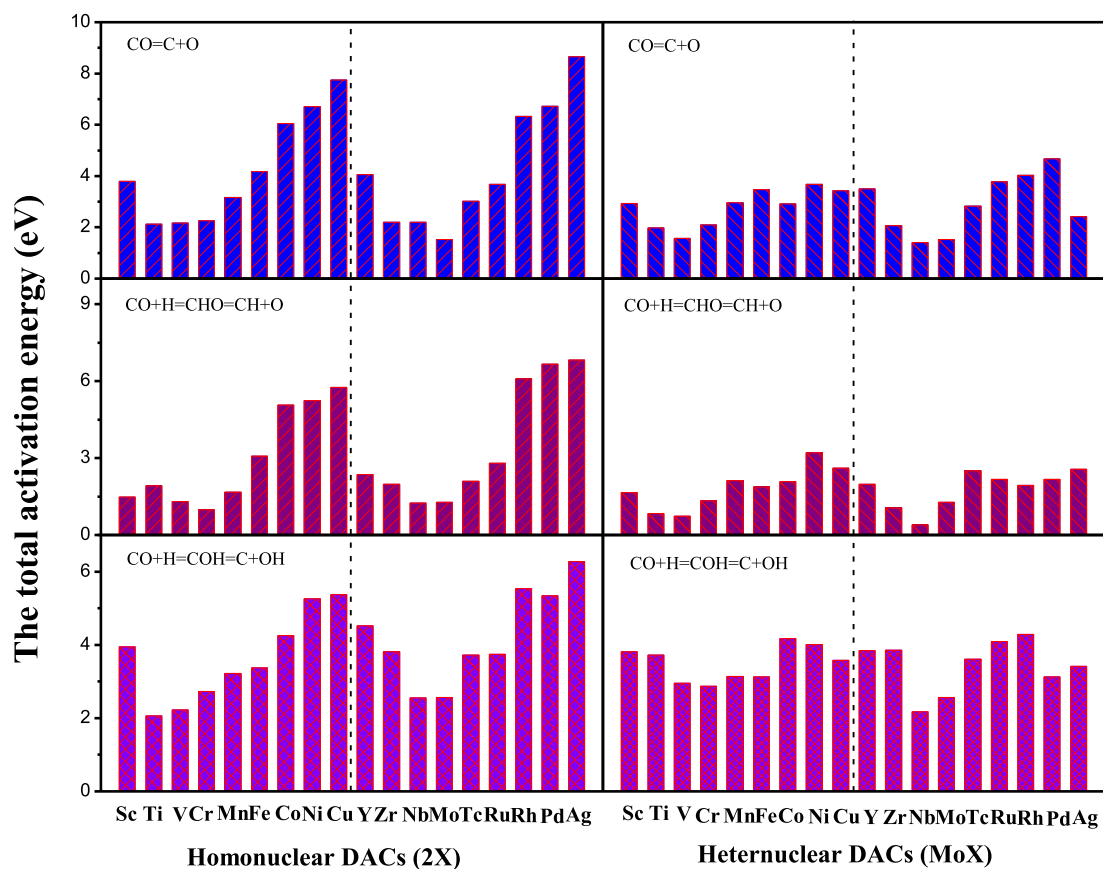


Fig. 3. The total activation energy (eV) of three CO activation mechanism for the homonuclear and heteronuclear DACs.

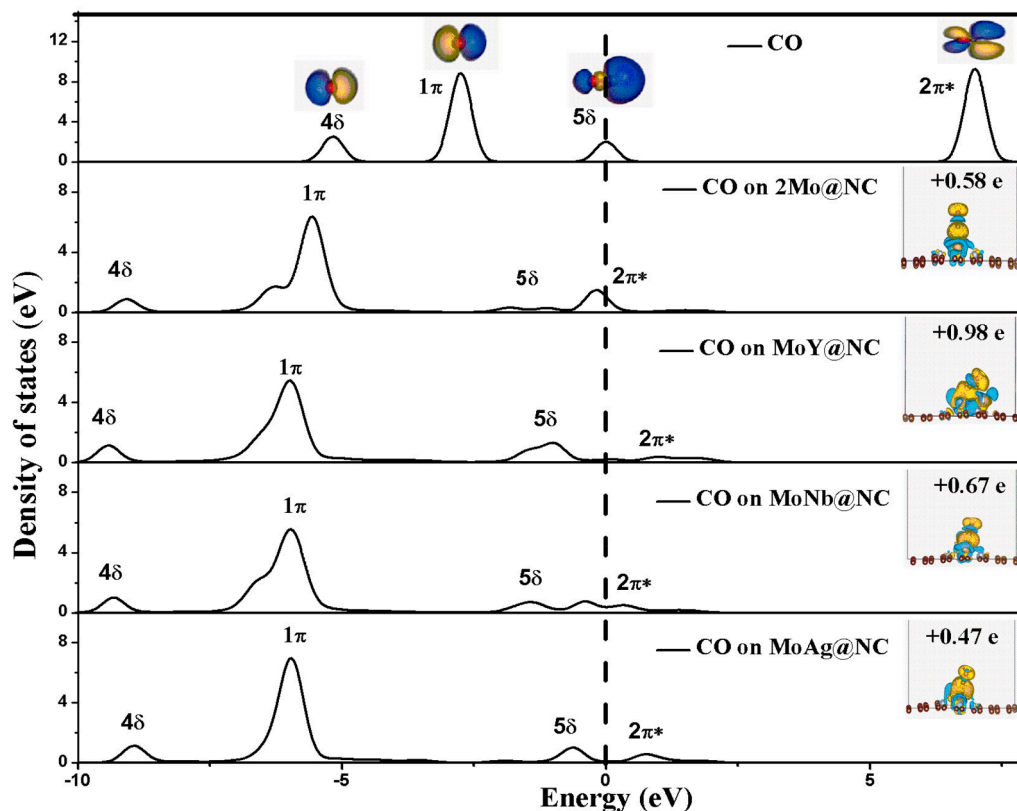


Fig. 4. The density of states (DOS) of CO, CO on 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts.

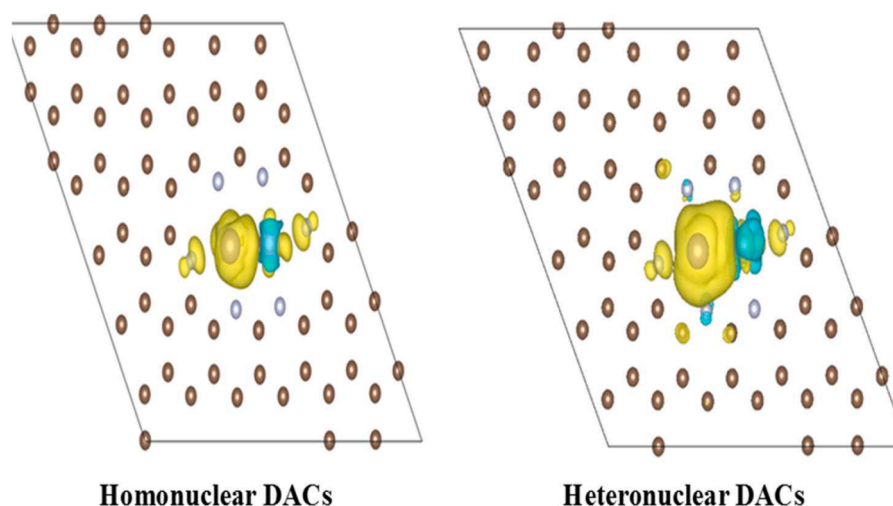


Fig. 5. The spin density for the homonuclear and heteronuclear DACs. yellow represents spin-up density and the blue-green represents spin-down density. (the isosurface value was $0.003 \text{ e } \text{\AA}^{-3}$).

it reveals that COH is difficult to generate through CO hydrogenation for all the 3d-TM heteronuclear DACs. The activation energy E_{a4} of $\text{COH} \rightarrow \text{C} + \text{OH}$ is in the range of 0.48 to 3.23 eV. In view of two continuous reaction, the MoCr@NC catalyst owns the highest catalytic performance for the CO activation through COH. The total activation energy E_{total} of MoCr@NC catalyst is 2.87 eV. However, the total activation energy E_{total} of MoCo@NC catalyst is 4.17 eV, and it is the greatest among them. For the 4d-TM heteronuclear DACs, the activation energy E_{a3} of $\text{CO} + \text{H} \rightarrow \text{COH}$ is in the range of 2.17 to 3.85 eV, it indicates that COH is also difficult to generate through CO hydrogenation for all the 4d-TM heteronuclear DACs, hydrogenation to CHO is the favorable pathway for carbon monoxide. The activation energy E_{a4} of $\text{COH} \rightarrow \text{C} + \text{OH}$ is in the range of 0.67–3.94 eV. In view of two continuous

reaction, MoNb@NC catalyst owns the highest catalytic performance for the CO activation through COH. The total activation energy E_{total} of MoNb@NC catalyst is 2.17 eV. However, the total activation energy E_{total} of MoRh@NC catalyst is 4.28 eV, which is the greatest among them.

Comprehensively considering the three mechanisms of CO activation of the homonuclear and heteronuclear DACs, as revealed in the Fig. 3, from IIIB to IB subgroup, whether 3d- or 4d-TM DACs, the activation energy shows an obvious tendency of first falling and then rising. And the *d*-orbital electron number of transition metals is increasing gradually from IIIB to IB subgroup. However, the unpaired *d*-orbital electron number (n_{ud}) of transition metals is first rising and then falling based on Hund's Rule, so it is concluded that the catalytic ability of the DACs for

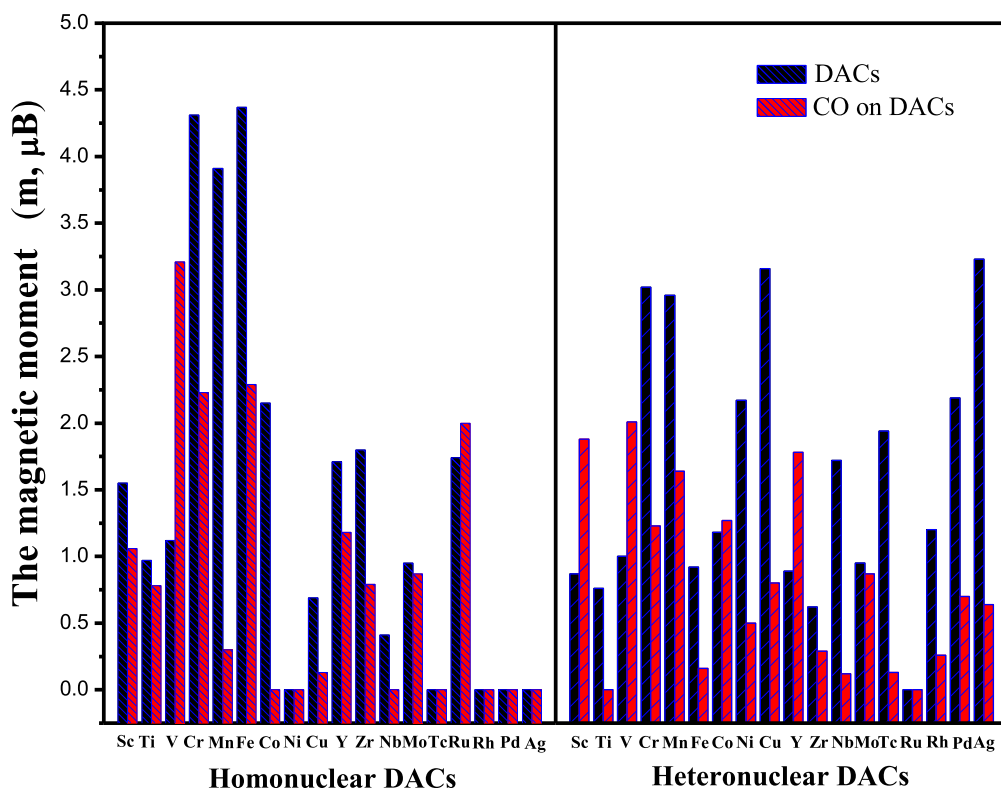


Fig. 6. The magnetic moments ($m, \mu\text{B}$) of the DACs and CO molecule adsorbed on the DACs.

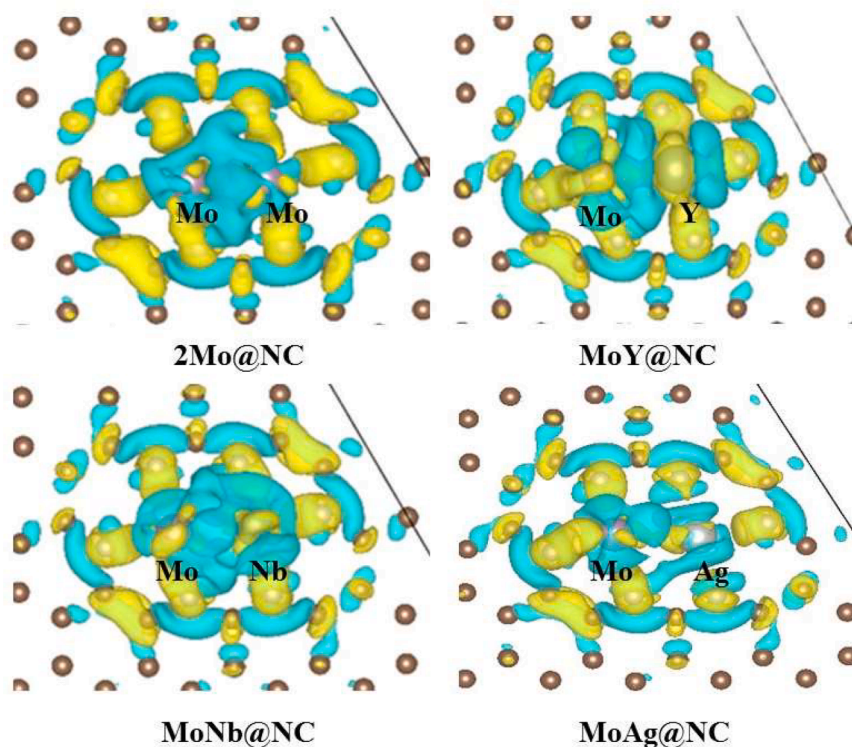


Fig. 7. The differential charge density of 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts. the isosurface value was $0.003 \text{ e } \text{\AA}^{-3}$ (yellow represent charge accumulation, and blue represent charge depletion).

CO activation associates with the unpaired d -orbital electron number (n_{ud}) of transition metals. The greater the unpaired d -orbital electron number (n_{ud}) of transition metals, the better the catalytic performance of the DACs.

Moreover, as revealed in the Fig. 3, for the same transition metals, the activation energy E_a of the heteronuclear DACs (MoX@NC) is significantly lower than that of homonuclear DACs (2X@NC). For example, the activation energy E_a of 2Cu@NC catalyst is 7.74 eV however, the activation energy E_a of MoCu@NC catalyst is only 3.43 eV for CO direct dissociation pathway. It attributes to synergistic effect between the Mo atom and other transition metals in dimer for the heteronuclear DACs. Considering the three mechanisms of CO activation, MoNb@NC catalyst possesses the best catalytic performance for all of the heteronuclear DACs.

3.5. Electronic property analysis

Considering the frontier molecular orbital (MO) theory, the electron interaction between the adsorbed Carbon monoxide and the transition atoms results from electronic donation and back-donation via the 5σ and $2\pi^*$ MOs of CO molecule. 5σ MO is the highest occupied molecular orbital (HOMO) of CO, and $2\pi^*$ MO is the lowest unoccupied molecular orbital (LUMO) of CO. Accordingly, the density of states (DOS) of CO and CO adsorbed on DACs have been considered (2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts have been selected and displayed in Fig. 4). The peak in the Fermi level of CO denotes 5σ MO (HOMO) of CO and the peak above the Fermi level of CO denotes the $2\pi^*$ MO (LUMO), which is in agreement with past report [47]. The $2\pi^*$ MO (LUMO) of CO presents antibonding state. However, when CO adsorbed on the 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts, the $2\pi^*$ MO (LUMO) of CO obviously move towards the Fermi level, even below the Fermi level. And it reveals that the adsorption system becomes more stable when CO adsorbed on the homonuclear and heteronuclear DACs. The Bader charge shows that CO molecular accept plenty of electrons from DACs (+0.58 e , +0.98 e , +0.67 e and +0.47 e correspond

to 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts), it indicates that the back-donation electron from the transition metals to carbon monoxide plays significant role on CO activation. The electron interaction between the CO and the DACs result in the formation of metal-carbon bond on transition metal surface, which weaken carbon-oxygen bonds and lead to the bond rupture.

The unpaired d -orbital electron relates with the ferromagnetism of DACs, therefore, the spin density of the homonuclear and heteronuclear DACs have been calculated (as displayed in the Fig. 5). For the homonuclear and heteronuclear DACs, the spin density majorly centers on the dimer formed by transition metals, which is in consistent with the research in the past report [48,49]. Meanwhile, the magnetic moments (m , μB) of the DACs and CO adsorbed on DACs have been analyzed. As displayed in Fig. 6, the CO adsorption bring about an obvious variation of the magnetic moment of DACs. When CO molecule adsorbed on the homonuclear and heteronuclear DACs, the magnetic moments diminish for the great majority of DACs, and it indicates that the electronic transfers from transition metals to CO molecule majorly occurs on the unpaired d -orbital electron of transition metal, and it results in a decrease of magnetic moment. And the unpaired d -orbital electron plays significant role on the back-donation electron from the transition metals to Carbon monoxide.

The differential charge density of the homonuclear and heteronuclear DACs have been discussed, similarly, 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts are selected and displayed in Fig. 7. It is deduced that there is obvious electronic transfer between the graphene-supporter and the dimer for the homonuclear and heteronuclear DACs.

The coordination environment is the main influencing factor for the catalytic performance of SACs and DACs, and the coordination atom [50] and coordination number [51] determine the coordination environment of SACs and DACs. Active center (transition metal atom or dimer) is encompassed by the coordination atoms, and it constitutes the local coordination environment for the SACs and DACs. Hence, the Bader charge mean value of the coordination atoms for the homonuclear

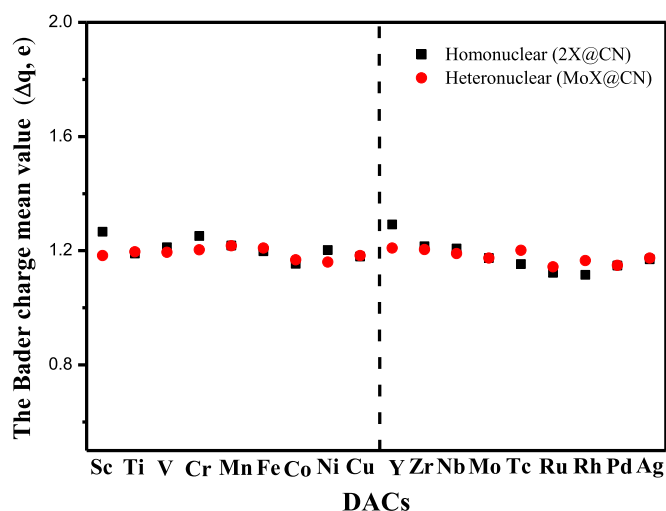


Fig. 8. The Bader charge mean value of the coordination atom for the homonuclear and heteronuclear DACs, the positive value represents the acquisition of electrons.

and heteronuclear DACs have been discussed. The Bader charges of the coordination atoms (N1; N2; N3; N4; N5; N6) of the homonuclear and heteronuclear DACs are listed in Table S7. The Bader charge mean value of the coordination atoms for the homonuclear DACs is in the range of +1.12 to +1.27 *e* (as revealed in the Fig. 8), and the difference value is only 0.15 *e*. Moreover, the Bader charges mean value of the coordination atoms for the heteronuclear DACs is in the range of +1.14 to +1.22 *e*, and the difference value is only 0.08 *e*. Such small variation indicates that the coordination atoms offer similar local coordination environment for the homonuclear and heteronuclear DACs.

The Bader charge of transition metal on the homonuclear and heteronuclear DACs also have been studied (as presented in Fig. 9). For the homonuclear DACs (2X@NC), from IIIB to IB subgroup, the Bader charge (Δq) of transition metal shows the decreasing trend, probably it

is caused by the weakened metallic property of transition metal. Compared to homonuclear DACs, the Bader charge (Δq) of transition metal enhances for the heteronuclear DACs (MoX@NC). The Bader charge (Δq) of early transition metals (Sc, Y, Ti, Zr, V and Nb) of the heteronuclear DACs (MoX@NC) become more than that of the homonuclear DACs, conversely, the Bader charge (Δq) of Mo atom (in MoX@NC) become less than that of homonuclear DACs; The Bader charge (Δq) of late transition metals (Fe, Ru, Co, Rh, Ni, Pd, Cu and Ag) of the heteronuclear DACs (MoX@NC) become less than that of homonuclear DACs, conversely, the Bader charge (Δq) of Mo atom (in MoX@NC) become more than that of homonuclear DACs. It clearly indicates that electron transfer exists between Mo atom and other transition metals for the heteronuclear DACs, it attributes to the diversity of metallic property of different transition metal. In other word, it has a strong electronic interaction between Mo atom and other transition metals for the heteronuclear DACs.

Because of the interaction of the *d*-orbital electron of transition metals, the metallic bond has been formed between transition metals of dimer on the homonuclear and heteronuclear DACs, and the metal-metal bonding strength can be explained by metallic bond length of dimer. Accordingly, the projected crystal orbital Hamilton population (*-pCOHP*) of 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts are calculated and displayed in Fig. 10. Antibonding characters of electronic states is denoted by negative values, and bonding characters of electronic states is denoted by positive value. Whether homonuclear or heteronuclear DACs, there is a large amount of electron located below the Fermi level and present a bonding state between two transition atoms of dimer. The integrated-COHP (ICOHP) could be used to illustrate the bonding strength [52], and the ICOHP value of 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts are -4.49, -4.89, -6.95 and -1.12, respectively. It displays that the metal-metal bonding is formed and there is strong electronic interaction between the two transition metals of dimer in the homonuclear and heteronuclear DACs. Similarly, the metallic bond length of dimer on the homonuclear and heteronuclear DACs have been discussed. As displayed in Fig. 11, the metallic bond length of dimer on the heteronuclear DACs (MoX@NC) is shorter than that of homonuclear DACs (2X@NC) for the great majority

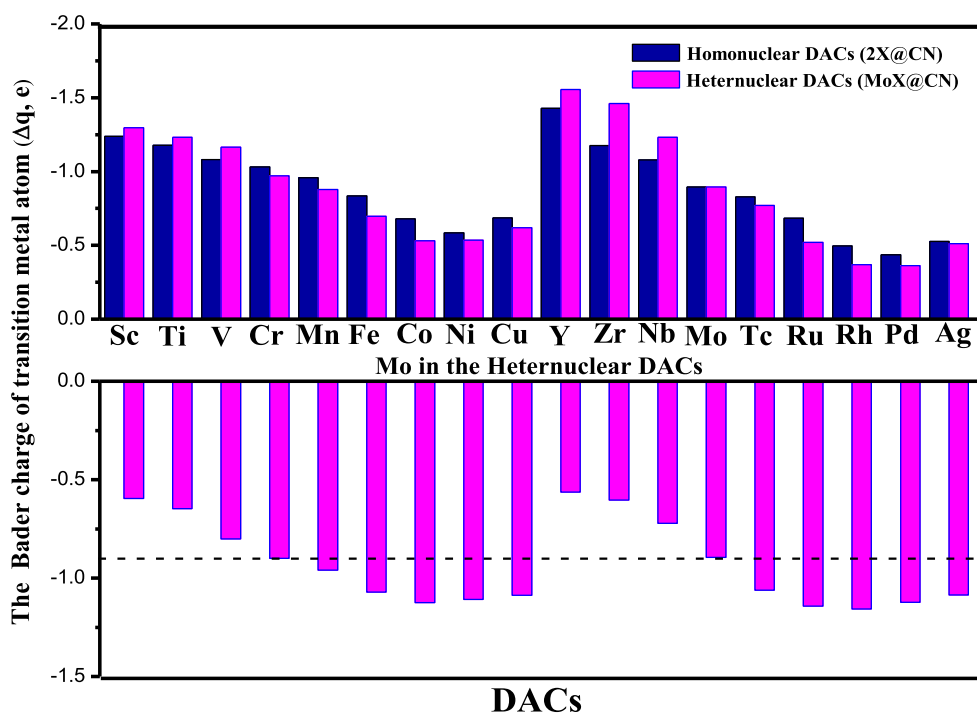


Fig. 9. The Bader charge of transition metal on the homonuclear and heteronuclear DACs, the negative values represent the loss of electrons.

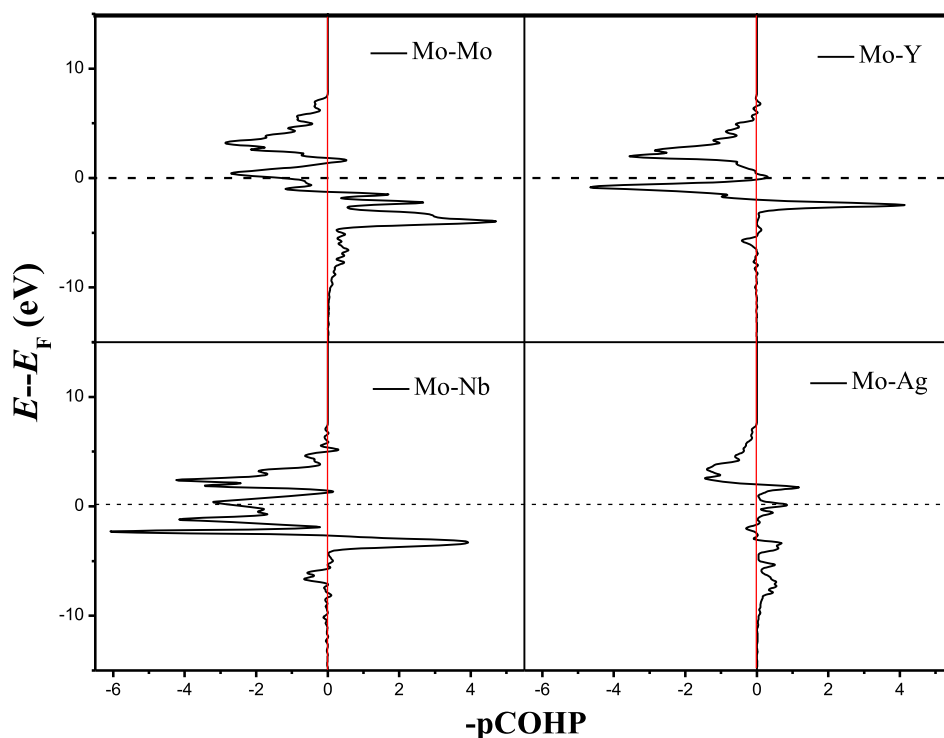


Fig. 10. The -pCOHP of 2Mo@NC, MoY@NC, MoNb@NC and MoAg@NC catalysts, and the Fermi level is set at 0 eV.

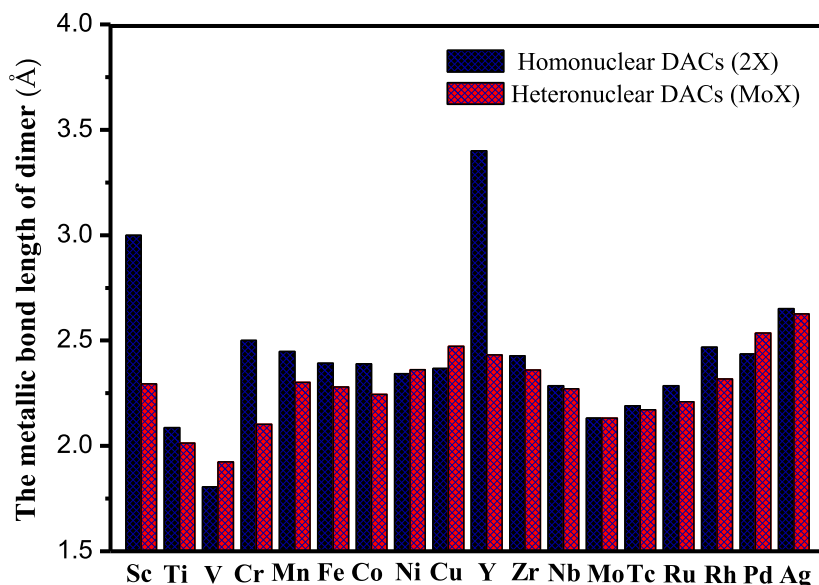


Fig. 11. The metallic bond length of dimer on the homonuclear and heteronuclear DACs.

of the DACs, especially for the MoSc@NC and MoY@NC catalysts. It indicates that compared to homonuclear DACs, the metal-metal bonding strength of dimer on the heteronuclear DACs is enhanced. Therefore, the synergistic effect of the heteronuclear DACs originates from the enhancement of the intermetallic interaction of dimer on DACs, and the synergistic effect enables the heteronuclear DACs to possess a higher performance than homonuclear DACs.

4. Conclusions

The Carbon monoxide (CO) dissociation on the homonuclear and

heteronuclear DACs has been extensively discussed by DFT calculation. In view of the activation energy (E_a) of CO activation, it concludes that the 2Mo@NC catalyst is the prime candidate of all the homonuclear DACs. The catalytic performance for CO activation closely relates with the unpaired d -orbital electron of transition metal on the homonuclear DACs. Based on the activation energy (E_a) of CO activation on the heteronuclear DACs, it displays that the heteronuclear DACs (MoX@NC) owns higher catalytic ability than homonuclear DACs (2X@NC), and it attributes to synergistic effect of dimer on the heteronuclear DACs. The Bader charge indicates that the coordination atoms offer similar local coordination environment for the homonuclear and heteronuclear

DACs. However, compared to homonuclear DACs, the electron transfer between the transition metals of dimer on the heteronuclear DACs is enhanced, and it has a strong electronic interaction between Mo atom and other transition metals for the heteronuclear DACs. Meanwhile, the metallic bond length of heteronuclear DACs (MoX@NC) is shorter than that of homonuclear DACs (2X@NC). The synergistic effect of the heteronuclear DACs originates from the enhancement of the intermetallic interaction between the transition metals of dimer. On the other hand, other transition metal can be used to improve the stability performance of 2Mo@NC catalyst. This work can offer theoretical strategies for the application of DACs on heterogeneous catalysis.

CRedit authorship contribution statement

Guorong Jia: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Lixia Ling:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Riguang Zhang:** Methodology, Funding acquisition. **Baojun Wang:** Supervision, Software, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.mcat.2024.114005](https://doi.org/10.1016/j.mcat.2024.114005).

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